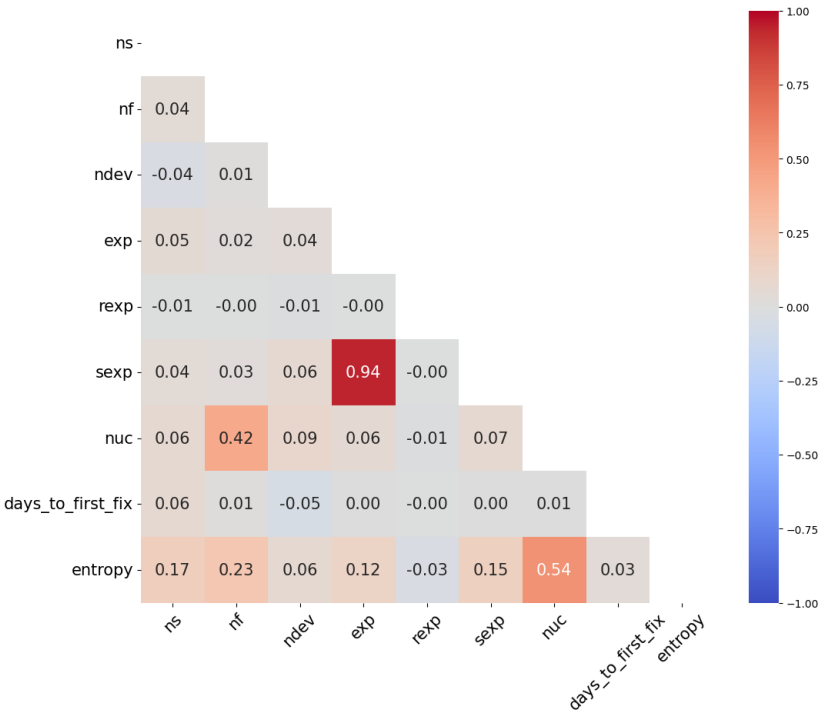


Figure A1: Heat map of correlation coefficients of Rust dataset

=0pt

Appendix

A. Dataset Description

A heat map is a data visualisation that represents different data intensities within a dataset using colour gradients. It is often used to identify correlations, patterns, and trends in data, making it easier to display complex information quickly. It conveys potential relationships between the variables shown on the x- and y-axes by using different colors, or varying shades of the same color, to indicate various values. Darker hues or shades typically indicate larger or more quantities of the values shown in the heat map. The reddish color represents the positive correlation, and the bluish color represents the negative correlation among the variables.We will also assess distribution shapes, variability, and central tendencies using descriptive statistics. It also enable us to identify outliers and unusual values. In the below sections, we have visualized each software projects using correlation metrics and descriptive statistics.

A1. Rust

The heat map of the correlation coefficient among the variables of our interest for the Rust data set is shown in Fig. A1.

Table A1 shows the statistics that can help to understand the distribution and variability of the different variables in the Rust dataset.

A2. TensorFlow

The heat map of the correlation coefficient among the variables of our interest for the Tensorflow data set is shown in Fig. A2.

Table A2 shows the statistics that can help to understand the distribution and variability of the different variables in the Tensorflow dataset.

Table A1
Descriptive statistics of the model variables for rust

Variable	Mean	Median	SD	Confidence Interval
ndev	348.79	304.00	270.09	(346.85, 350.74)
ns	1.03	1.00	0.21	(1.03, 1.04)
nf	6.43	2.00	92.97	(5.77, 7.11)
rexp	74.79	1.13	1321.50	(65.27, 84.32)
sexp	3835.32	673.5	6940.66	(3785.27, 3885.37)
entropy	0.8812	0.2422	1.2184	(0.87, 0.89)
days_to_first_fix	3.46	0.00	47.83	(3.12, 3.81)
exp	4255.27	889.25	7357.31	(4202.22, 4308.33)

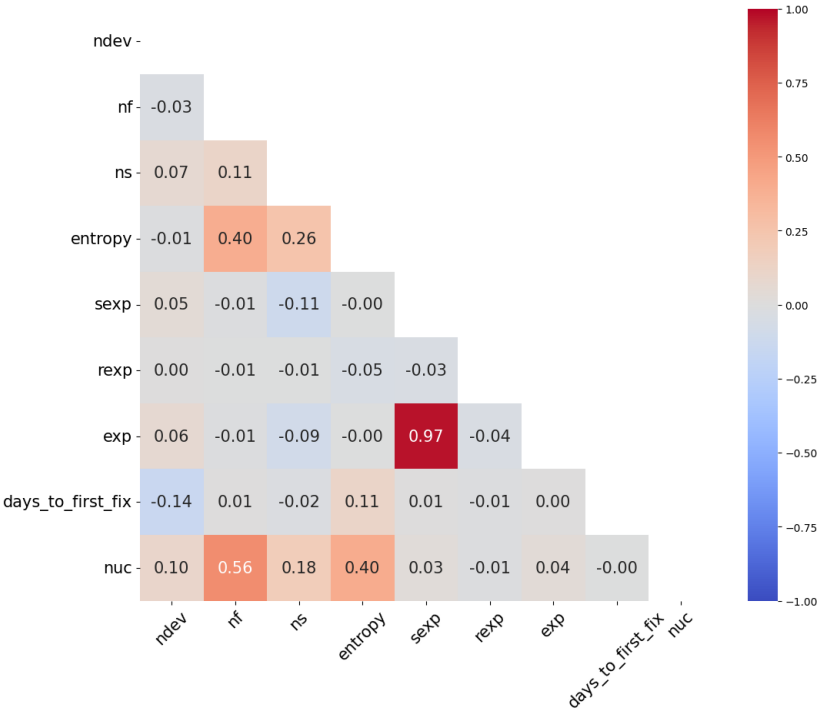


Figure A2: Heat map of correlation coefficients of Tensorflow dataset

A3. Rails

The heat map of the correlation coefficient among the variables of our interest for the Rails data set is shown in Fig. A3.

Table A3 shows the statistics that can help to understand the distribution and variability of the different variables in the Rails dataset.

A4. VSCode

The heat map of the correlation coefficient among the variables of our interest for the VSCode data set is shown in Fig. A4.

Table A4 shows the statistics that can help to understand the distribution and variability of the different variables in the VSCode dataset.

A5. WP-Calypso

The heat map of the correlation coefficient among the variables of our interest for the WP-Calypso data set is shown in Fig. A5.

Table A2
Descriptive statistics of the model variables for Tensorflow

Variable	Mean	Median	SD	Confidence Interval
ndev	268.72	285.00	140.67	(267.65, 269.81)
ns	1.07	1.00	0.36	(1.07, 1.08)
nf	5.23	2.00	35.58	(4.96, 5.51)
rexp	17.61	1.13	246.65	(15.72, 19.51)
sexp	20180.28	596.5	34515.36	(19915.01, 20445.56)
entropy	0.8850	0.6665	1.0911	(0.88, 0.89)
days_to_first_fix	24.60	0.00	87.73	(23.93, 25.28)
exp	21150.18	780.75	35330.65	(20878.64, 21421.73)

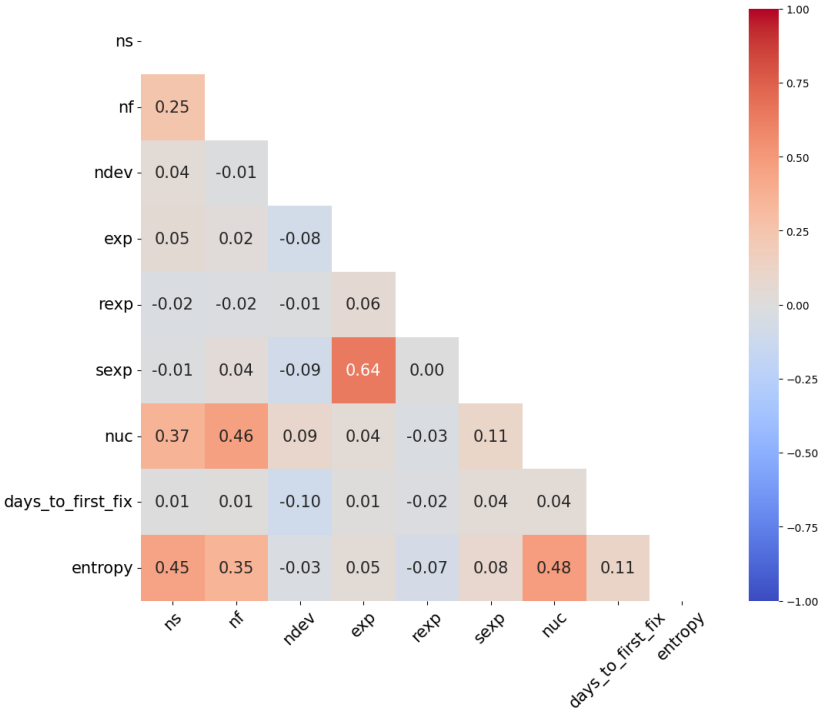


Figure A3: Heat map of correlation coefficients of Rails dataset

Table A5 shows the statistics that can help to understand the distribution and variability of the different variables in the WP-Calypso dataset.

A6. Camel

The heat map of the correlation coefficient among the variables of our interest for the Camel data set is shown in Fig. A6.

Table A6 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

A7. CoreFX

The heat map of the correlation coefficients among the variables of interest in the CoreFX dataset is shown in Fig. A7.

Table A7 shows the statistics that can help to understand the distribution and variability of the different variables in the CoreFX dataset.

Table A3
Descriptive statistics of the model variables for Rails

Variable	Mean	Median	SD	Confidence Interval
ndev	252.61	232.00	169.90	(251.23, 254.00)
ns	1.15	1.00	0.66	(1.15, 1.16)
nf	2.76	1.00	8.20	(2.70, 2.84)
rexp	65.09	1.119	549.86	(60.62, 69.57)
sexp	428.39	77.0	746.29	(422.32, 434.48)
entropy	0.6316	0.00	0.8885	(0.63, 0.64)
days_to_first_fix	106.44	0.00	352.34	(103.57, 109.31)
exp	1713.74	450.0	2613.08	(1692.47, 1735.02)

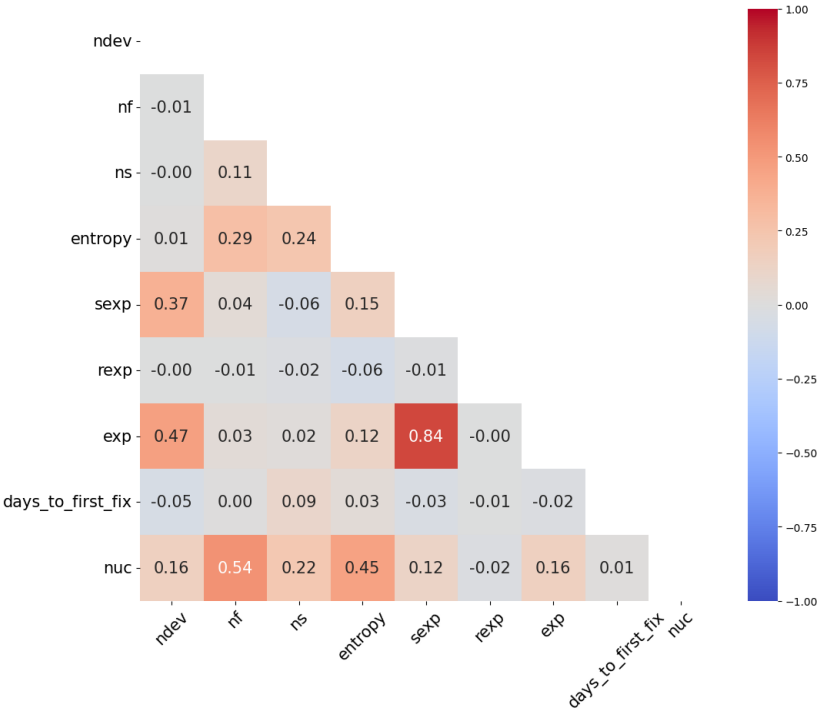


Figure A4: Heat map of correlation coefficients of VSCode dataset

A8. Django

The heat map of the correlation coefficients among the variables of interest in the Django dataset is shown in Fig. A8.

Table A8 shows the statistics that can help to understand the distribution and variability of the different variables in the Django dataset.

A9. Nova

The heat map of the correlation coefficients among the variables of interest in the Nova dataset is shown in Fig. A9.

Table A9 shows the statistics that can help to understand the distribution and variability of the different variables in the Nova dataset.

A10. JGroup

The heat map of the correlation coefficients among the variables of interest in the JGroup data set is shown in Fig. A10.

Table A4
Descriptive statistics of the model variables for VSCode

Variable	Mean	Median	SD	Confidence Interval
ndev	60.60	63.00	27.40	(60.37, 60.84)
ns	1.05	1.00	0.26	(1.05, 1.05)
nf	4.59	1.00	55.96	(4.11, 5.07)
rexp	52.53	1.40	422.99	(48.89, 56.17)
sexp	4739.20	2312.5	5942.17	(4688.06, 4790.36)
entropy	0.7228	0.00	1.0209	(0.71, 0.73)
days_to_first_fix	2.67	0.00	32.59	(2.40, 2.96)
exp	6601.52	4540.75	6804.46	(6542.96, 6660.10)

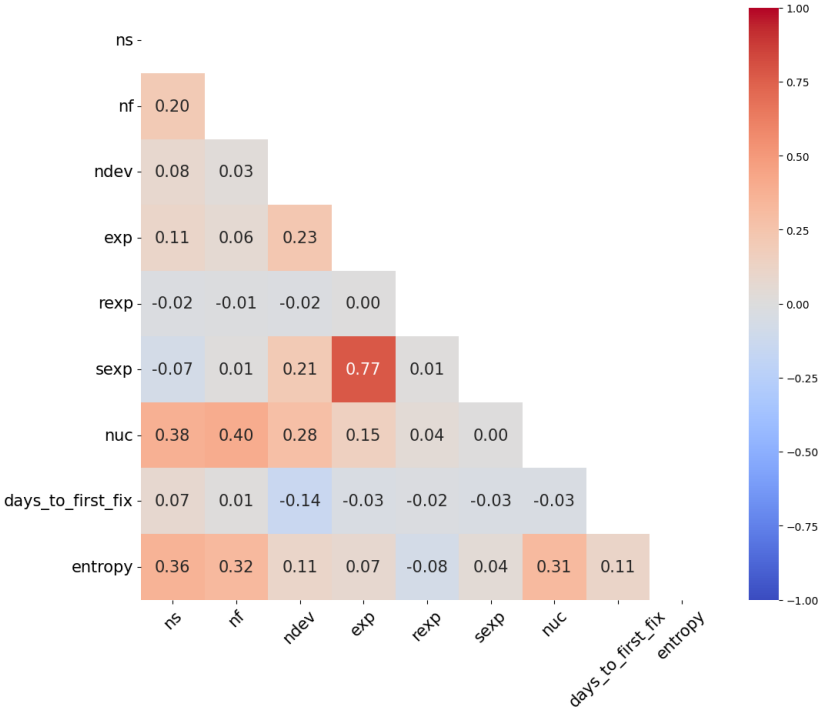


Figure A5: Heat map of correlation coefficients of WP-Calypso dataset

Table A10 shows the statistics that can help to understand the distribution and variability of the different variables in the JGroup dataset.

A11. Fabric 8

The heat map of the correlation coefficient among the variables of our interest for the fabric data set is shown in Fig. A11.

Table A11 shows the statistics that can help to understand the distribution and variability of the different variables in the fabric dataset.

A12. Broadleaf

The heat map of the correlation coefficient among the variables of our interest for the Broadleaf data set is shown in Fig. A12.

Table A12 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

Table A5
Descriptive statistics of the model variables for WP-Calypso

Variable	Mean	Median	SD	Confidence Interval
ndev	138.94	146.00	75.89	(138.11, 139.79)
ns	1.11	1.00	0.39	(1.11, 1.12)
nf	3.85	2.00	27.52	(3.55, 4.16)
rexp	48.91	1.11	337.92	(45.16, 52.66)
sexp	507.74	206.0	778.86	(499.11, 516.39)
entropy	0.77	0.4262	0.99	(0.76, 0.79)
days_to_first_fix	45.27	0.00	154.18	(43.56, 46.98)
exp	701.74	337.0	998.26	(690.67, 712.83)

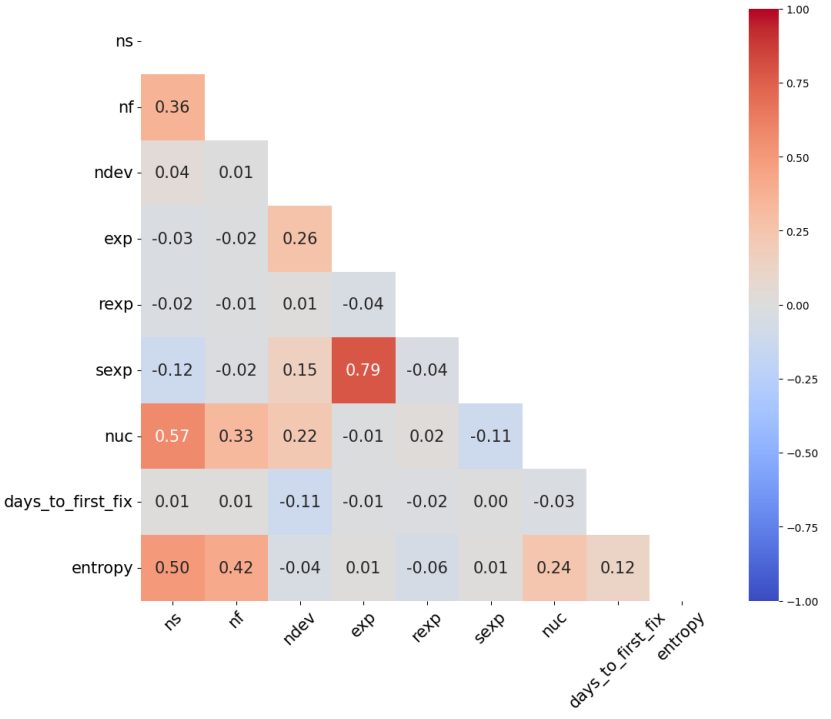


Figure A6: Heat map of correlation coefficients of Camel dataset

A13. Brackets

The heat map of the correlation coefficient among the variables of our interest for the Brackets data set is shown in Fig. A13.

Table A13 shows the statistics that can help to understand the distribution and variability of the different variables in the dataset.

B. Violin Plot

A violin plot combines a box plot with a mirrored kernel density estimate (KDE) of a variable's distribution, where wider sections indicate higher observation density.

We use violin plots throughout this section to compare each variable's distribution across all 13 projects — for example, whether bug resolution times cluster around a particular value or spread widely. Several violin plots are provided in Figs.B.1, B.2, B.3 and B.4, which show density extending into negative values for variables that are non-negative by definition (e.g., $ndev$, exp, $days_to_first_fix$ cannot be less than zero).

Table A6
Descriptive statistics of the model variables for Camel

Variable	Mean	Median	SD	Confidence Interval
ndev	47.31	31.00	44.80	(46.81, 47.82)
ns	1.28	1.00	0.83	(1.27, 1.28)
nf	6.77	2.00	41.32	(6.30, 7.24)
rexp	63.45	1.06	682.88	(55.75, 71.15)
sexp	5149.49	844.00	8191.84	(5057.15, 5241.85)
entropy	1.0853	0.8136	1.3254	(1.07, 1.10)
days_to_first_fix	48.74	0.00	225.34	(46.20, 51.28)
exp	15312.83	5702.50	19512.00	(15092.86, 15532.80)

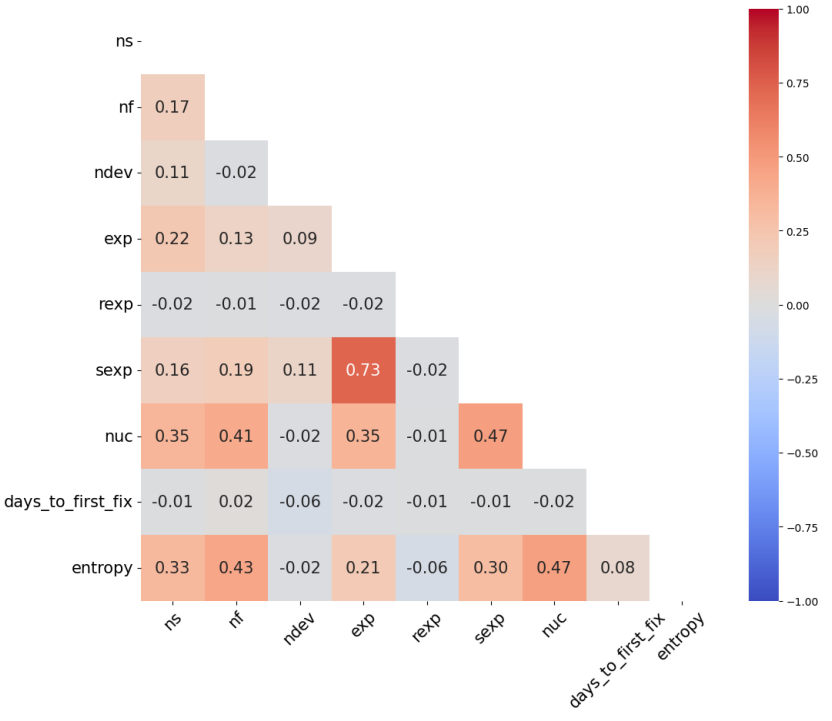


Figure A7: Heat map of correlation coefficients of CoreFX dataset

This visualisation artefact arises from kernel density estimation, which uses smooth, unbounded kernels that can extend density estimates beyond the data's true support; it does not indicate negative values in the underlying dataset. All raw values for these variables are greater than 0. Readers should interpret the tails of these plots qualitatively (as indicating low-density regions) rather than literally.

Table A7
Descriptive statistics of the model variables for CoreFX

Variable	Mean	Median	SD	Confidence Interval
ndev	193.68	188.00	109.81	(192.37, 195.01)
ns	1.13	1.00	0.41	(1.13, 1.14)
nf	11.71	2.00	78.34	(10.77, 12.66)
rexp	25.11	1.07	228.37	(22.37, 27.85)
sexp	3538.41	440.00	8407.69	(3437.42, 3639.41)
entropy	1.17	0.83	1.52	(1.16, 1.19)
days_to_first_fix	13.53	0.00	85.95	((12.50, 14.57)
exp	4613.17	788.00	11077.52	(4480.12, 4746.24)

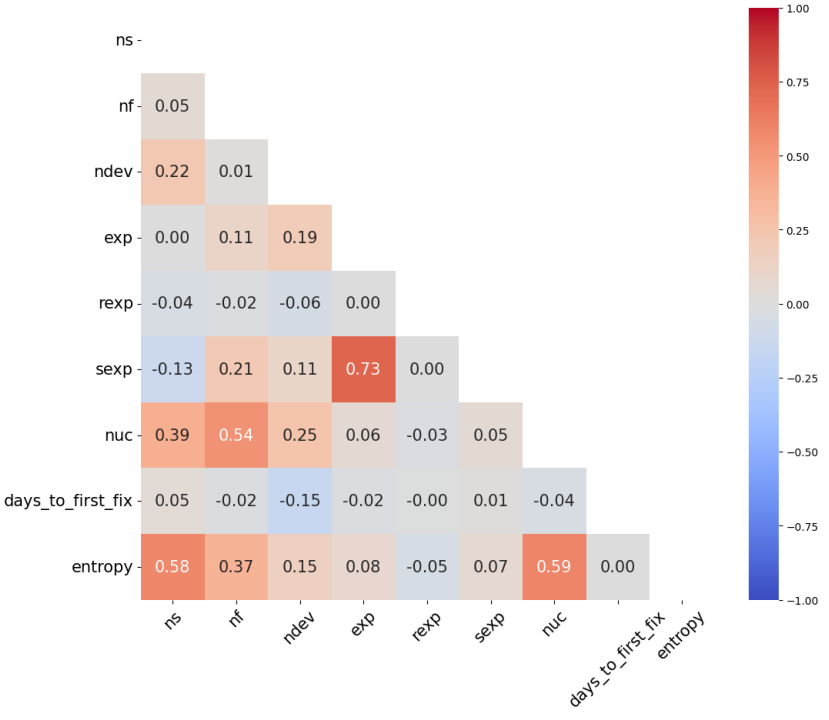


Figure A8: Heat map of correlation coefficients of Django dataset

B1. Violin plot for developer count
B2. Violin plot for file count
B3. Violin plot for overall experience
B4. Violin plot for resolution time

C. Summary statistics

C1. Summary statistics of developer count across all 13 project datasets
C2. Summary statistics of entropy across all 13 project datasets
C3. Summary statistics of *subsystem experience* across all 13 project datasets.
C4. Summary statistics of *recent experience* across all 13 project datasets.

D. Falsification of the causal graph

A causal graph is a set of claims about the data, and some of them can be checked. Every absent edge implies a conditional independence: under the local Markov condition, each variable should be independent of its non-descendants once its parents are known. If the data contradict many of these implications, the graph cannot be the

Table A8
Descriptive statistics of the model variables for Django

Variable	Mean	Median	SD	Confidence Interval
ndev	77.25	35.00	81.19	(76.27, 78.23)
ns	1.45	1.00	0.72	(1.45, 1.46)
nf	4.37	2.00	33.82	(3.97, 4.79)
rexp	41.75	1.03	414.70	(36.74, 46.76)
sexp	802.20	268.00	1432.75	(784.91, 819.50)
entropy	0.7102	0.00	1.0183	(0.70, 0.72)
days_to_first_fix	153.80	0.00	377.60	(149.25, 158.36)
exp	2335.92	836.25	3360.53	(2295.35, 2376.49)

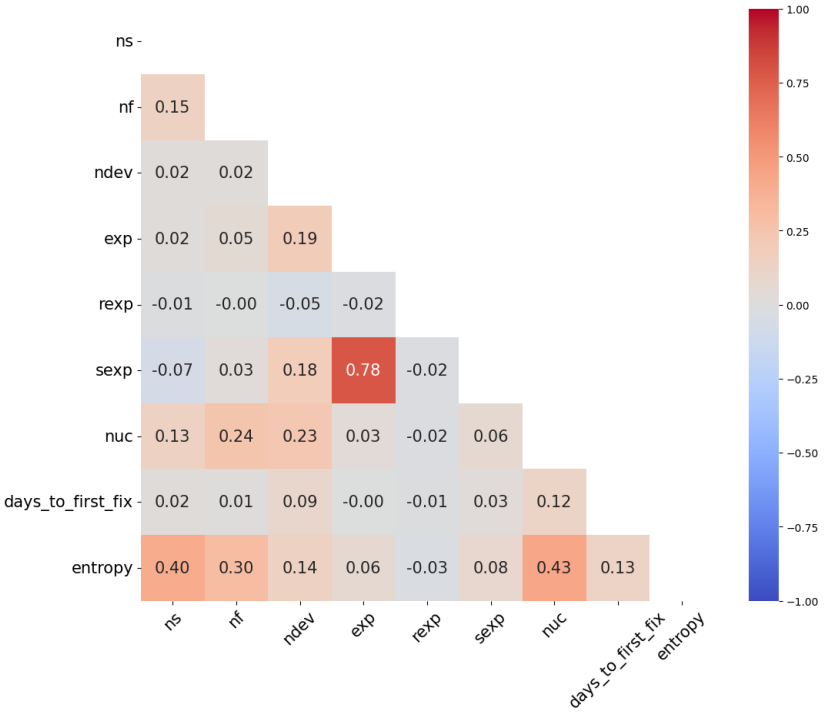


Figure A9: Heat map of correlation coefficients of Nova dataset

process that produced them, and estimates built on it are unreliable. This section reports how the graph of Fig. 2 fares on each of the 13 projects. Passing the test does not show that the graph is correct; it shows only that the data fail to contradict it.

Procedure. We used the permutation-based procedure of Eulig et al.[**?**], implemented in DoWhy's gcm.falsify module, in four steps.

1. **Test the graph's implications**. For each node, we tested its independence from every non-descendant given its parents, using a significance level of 0.05. The graph of Fig. 2 implies 66 such statements. We used a partial-correlation test on ranks (a Spearman partial correlation). It is robust to the heavy tails of la, lt and nuc and remained practical at our sample sizes, unlike kernel-based tests.
2. **Count violations**. We recorded the share of the 66 statements the data reject.
3. **Build a baseline**. A raw count of violations is hard to read on its own. With several thousand commits per project, a rank test detects even very weak dependences, so most graphs over these variables violate a large share of their implications. The procedure therefore compares the proposed graph with 1,000 graphs that keep its structure but randomly permute which variable sits at which node. Each permuted graph is tested in the same way.
4. **Compute two quantities.**

Table A9
Descriptive statistics of the model variables for Nova

Variable	Mean	Median	SD	Confidence Interval
ndev	168.82	149.00	118.28	(167.40, 170.25)
ns	1.14	1.00	0.44	(1.14, 1.15)
nf	4.49	2.00	36.56	(4.05, 4.94)
rexp	124.65	1.04	2619.02	(93.01, 156.30)
sexp	363.62	90.0	744.30	(354.63, 372.62)
entropy	0.8764	0.7219	1.0609	(0.86, 0.89)
days_to_first_fix	92.37	0.00	220.29	(89.71, 95.04)
exp	532.47	164.5	980.44	(520.62, 544.32)

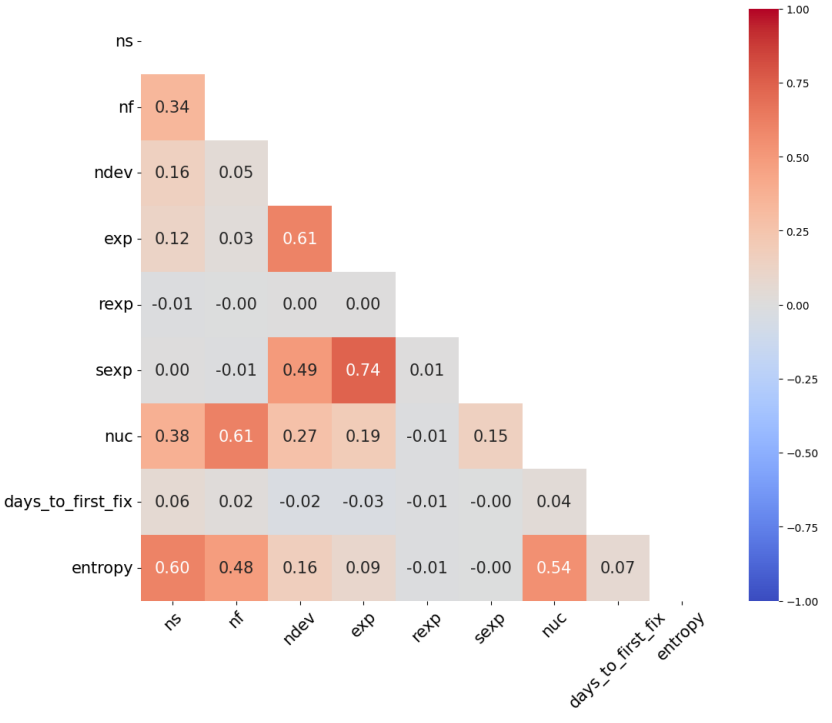


Figure A10: Heat map of correlation coefficients of JGroup dataset

- **p_LMC** is the share of permuted graphs that violate no greater share of their implications than the proposed graph does. A small value means the proposed graph fits the data better than almost all structurally identical alternatives. A large value means it does no better than a random relabelling. Following DoWhy's default, we reject the graph for a project when $p_LMC > 0.05$.
- **p_TPa** is the share of permuted graphs that fall in the proposed graph's Markov equivalence class. When it is small, the test is informative: the data could, in principle, distinguish the proposed graph from its permutations. A test that cannot do so would be uninformative, whatever its p_LMC.

To check that verdicts did not depend on the random permutations drawn, we repeated the test with three random seeds. Because every test is a partial correlation, all of them can be computed from one rank-correlation matrix per project. We used this to run the 39,000 permutations efficiently, and confirmed the verdicts with DoWhy's own implementation.

Results: Figures D1 and D2 show histograms of LMC violations for permuted DAGs and TPa violations, and each panel's legend shows the colour scheme.

Table A10
Descriptive statistics of the model variables for JGroup

Variable	Mean	Median	SD	Confidence Interval
ndev	10.83	11.00	7.07	(10.73, 10.94)
ns	1.17	1.00	0.48	(1.17, 1.18)
nf	2.51	1.00	13.09	(2.32, 2.70)
rexp	47.87	1.0039	1232.16	(29.88, 65.88)
sexp	3342.80	2539.0	3167.17	(3296.54, 3389.07)
entropy	0.3983	0.0	0.8194	(0.39, 0.41)
days_to_first_fix	73.14	0.00	312.84	(68.58, 77.72)
exp	8293.86	7846.0	6150.00	(8204.03, 8383.71)

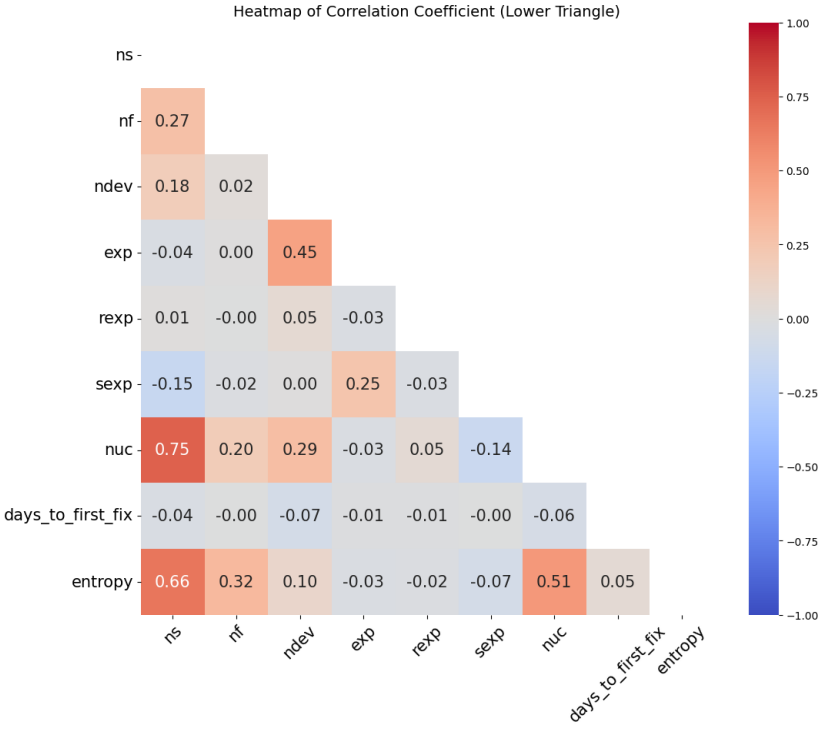


Figure A11: Heat map of correlation coefficients of Fabric 8 dataset

- **Informative everywhere.** No permuted graph fell in the proposed graph's equivalence class in any project ($p_{\text{TPa}} = 0.000$).
- **Not rejected in 11 projects.** Brackets, Camel, CoreFX, Django, Fabric8, Nova, Rails, Rust, TensorFlow, VSCode and WP-Calypso all have p_LMC between 0.001 and 0.04 across seeds. In each of them, the proposed graph violates fewer of its implications than at least 96% of its permutations.
- **Rejected in two projects.** For Broadleaf (p_LMC 0.24–0.29) and JGroup (0.22–0.24), the graph fits no better than a random relabelling of its variables.
- **Robust verdicts.** The two groups are far apart. Any cutoff between 0.04 and 0.31 yields the same verdicts, and all three seeds agree on every project.

Table A11
Descriptive statistics of the model variables for Fabric 8

Variable	Mean	Median	SD	Confidence Interval
ndev	32.10	31.00	16.72	(31.82, 32.40)
ns	1.53	1.00	1.79	(1.51, 1.57)
nf	9.46	2.00	84.56	(7.98, 10.95)
rexp	186.125	1.20	3249.38	(129.14, 243.10)
sexp	895.08	179.00	1469.96	(869.31, 920.86)
entropy	1.0616	0.2108	1.5323	(1.03, 1.09)
days_to_first_fix	25.58	0.00	97.95	(23.86, 27.30)
exp	6918.40	3954.00	8384.15	(6771.39, 7065.43)

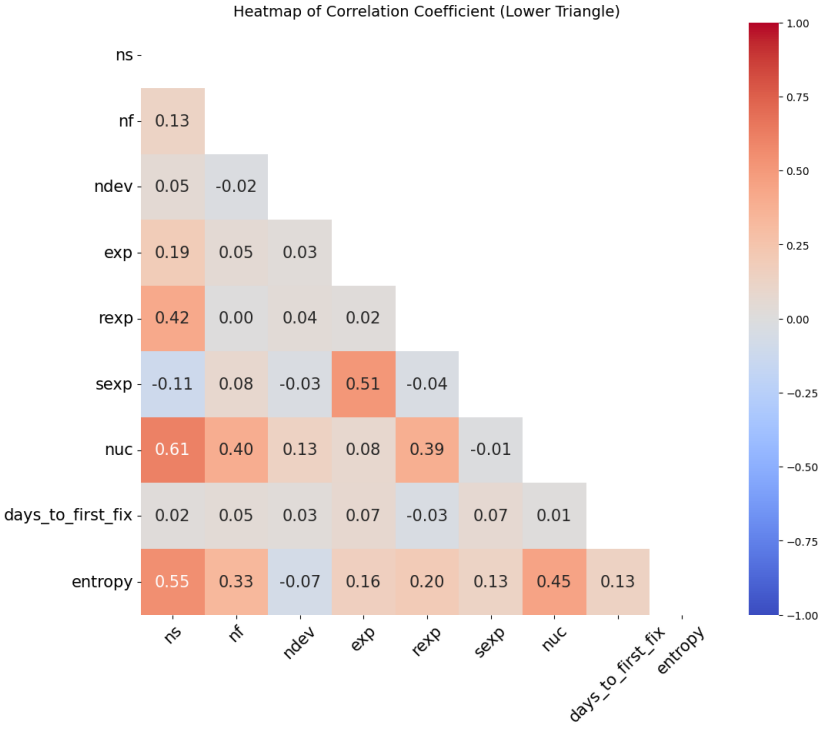


Figure A12: Heat map of correlation coefficients of Broadleaf dataset

E. Causal-effect estimates

E1. Causal effect of ndev on defect lifetimee

Table E1 summarises the causal inference results for developer count on defect lifetime across different software.

E2. Causal effect of entropy on defect lifetime

Table E2 summarises the causal inference results for the effect of *entropy* on defect lifetime across different software.

E3. Causal effect of sexp on defect lifetime

Table E3 summarises the causal inference results for subsystem experience on defect lifetime across software.

E4. Causal effect of rexp on defect lifetime

Table E4 summarises the causal inference results for rexp on days_to_first_fix.

Table A12
Descriptive statistics of the model variables for Broadleaf

Variable	Mean	Median	SD	Confidence Interval
ndev	21.07	24.00	11.00	(20.88, 21.27)
ns	1.33	1.00	0.96	(1.32, 1.35)
nf	9.77	2.00	102.34	(7.97, 11.58)
rexp	129.34	1.09	879.91	(113.81, 144.86)
sexp	1643.81	327.50	3920.65	(1574.62, 1713.01)
entropy	1.0388	0.5586	1.3546	(1.01, 1.06)
days_to_first_fix	48.78	0.00	202.03	(45.21, 52.34)
exp	4628.11	1205.00	7577.76	(4494.37, 4761.85)

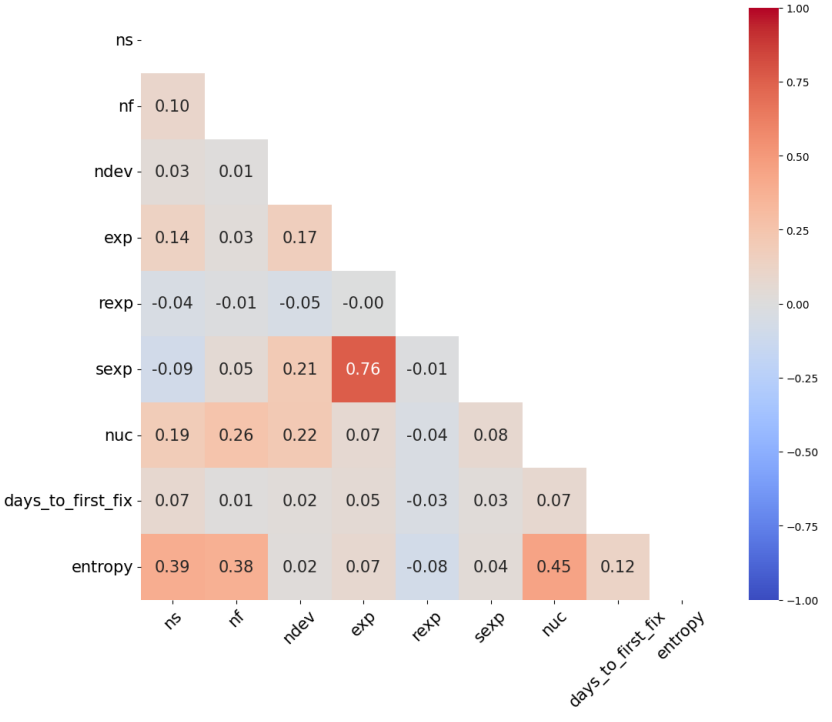


Figure A13: Heat map of correlation coefficients of Brackets dataset

References

[1] Eulig, E., Mastakouri, A.A., Blöbaum, P., Hardt, M., Janzing, D.: Toward falsifying causal graphs using a permutation-based test. In: Proceedings of the AAAI Conference on Artificial Intelligence, vol. 39, no. 25, pp. 26778–26786 (2025)

Table A13
Descriptive statistics of the model variables for Brackets

Variable	Mean	Median	SD	Confidence Interval
ndev	42.13	41.00	26.13	(41.66, 42.61)
ns	1.12	1.00	0.34	(1.11, 1.12)
nf	3.05	1.00	26.25	(2.58, 3.53)
rexp	43.76	1.61	264.21	(38.95, 48.57)
sexp	365.39	143.00	511.38	(356.09, 374.71)
days_to_first_fix	58.59	0.00	214.69	(54.69, 62.50)
entropy	0.4887	0.00	0.7824	(0.47, 0.50)
exp	668.61	246.00	1017.29	(650.10, 687.13)

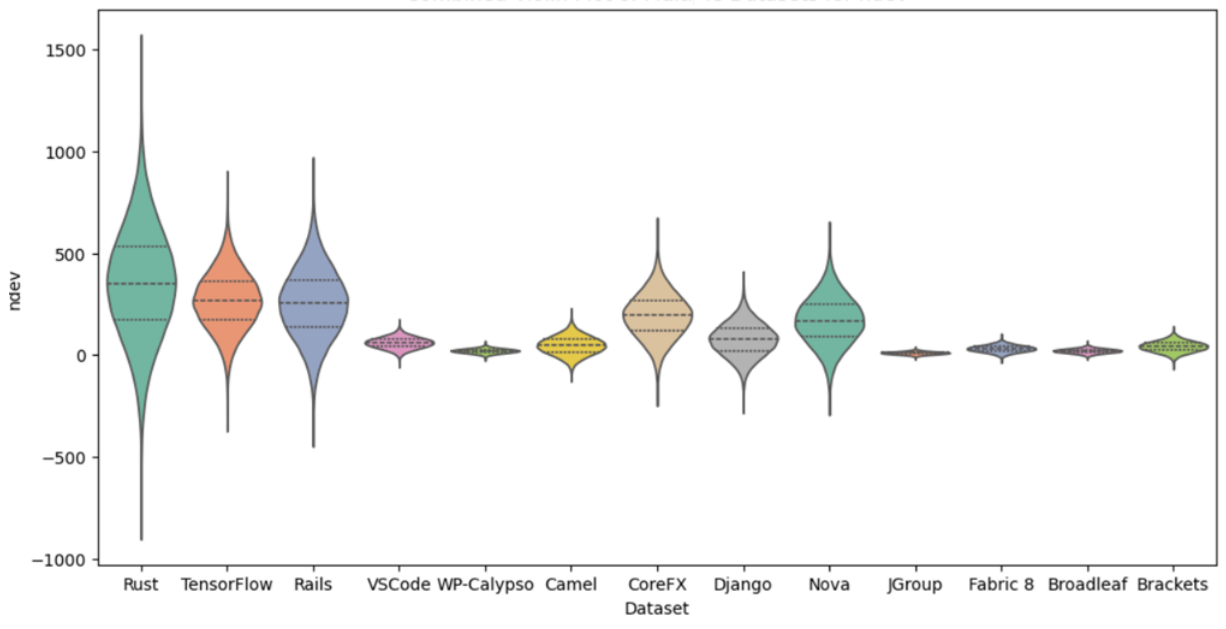


Figure B1: Violin Plot of Multiple Datasets for ndev

Project	Mean	Median	SD	Skew (Pearson's 2nd coeff.)
Brackets	42.13	41.00	26.13	Right (0.130)
Camel	47.31	31.00	44.80	Right (1.092)
CoreFX	193.68	188.00	109.81	Right (0.155)
Fabric 8	32.10	31.00	16.72	Right (0.197)
Rails	252.61	232.00	169.90	Right (0.364)
TensorFlow	268.72	285.00	140.67	Left (-0.347)
VSCode	60.60	63.00	27.40	Left (-0.263)
WP-Calypso	138.94	146.00	75.89	Left (-0.279)
Rust	348.79	304.00	270.09	Right (0.498)
Django	77.25	35.00	81.19	Right (1.561)
Nova	168.82	149.00	118.28	Right (0.503)
JGroup	10.83	11.00	7.07	Left (-0.072)
Broadleaf	21.07	24.00	11.00	Left (-0.799)

Table C1
Summary statistics of *developer count* across all 13 project datasets.

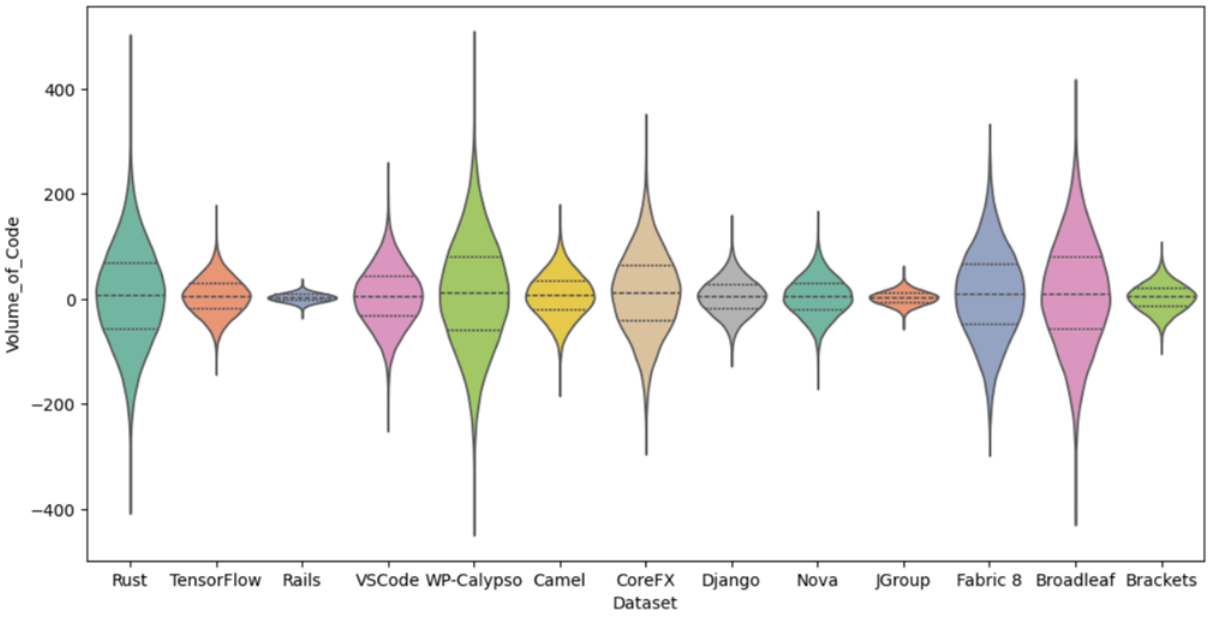


Figure B2: Violin Plot of Multiple Datasets for Volume_of_Code(= nf)

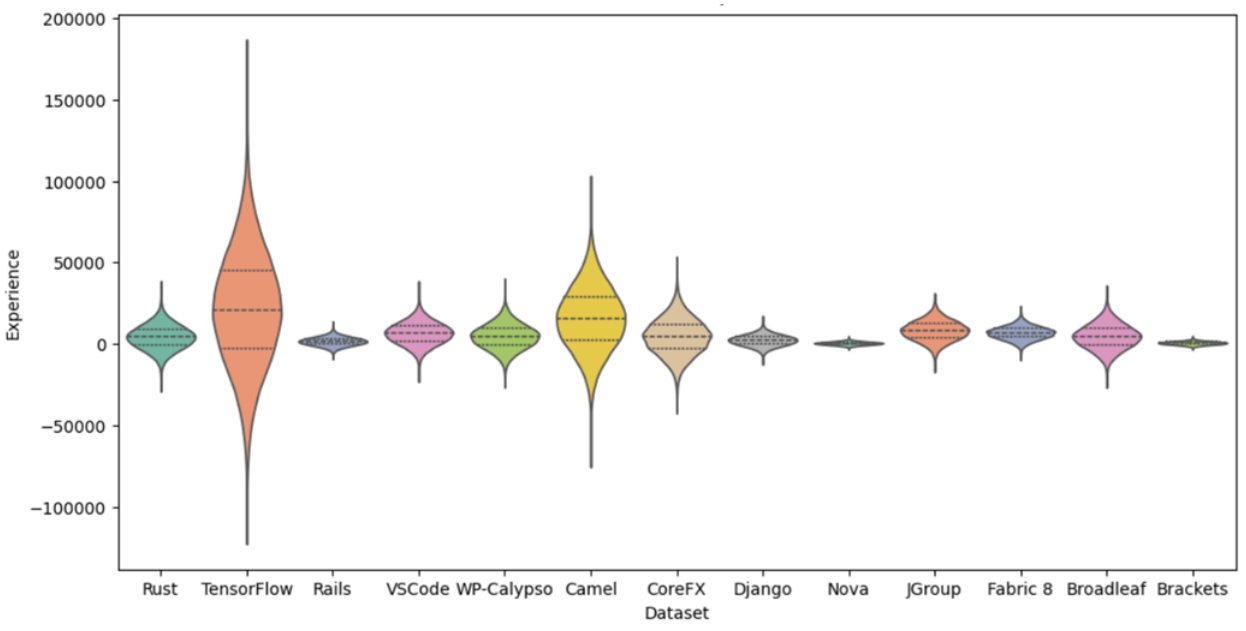


Figure B3: Violin Plot of Multiple Datasets for Experience (= exp)

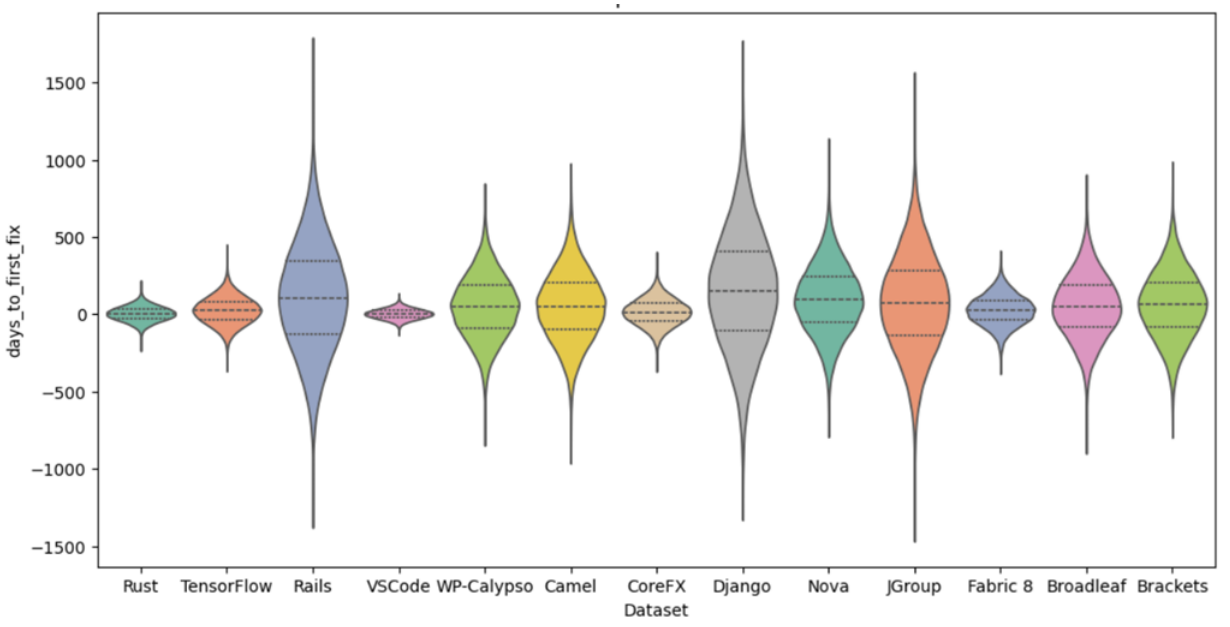


Figure B4: Violin Plot of Multiple Datasets for Bug Resolution Time

Project	Mean	Median	SD	Skew (Pearson's 2nd coeff.)
Brackets	0.4887	0.00	0.7824	Right (1.874)
Camel	1.0853	0.8136	1.3254	Right (0.615)
CoreFX	1.17	0.83	1.52	Right (0.671)
Fabric 8	1.0616	0.2108	1.5323	Right (1.666)
Rails	0.6316	0.00	0.8885	Right (2.133)
TensorFlow	0.8850	0.6665	1.0911	Right (0.601)
VSCode	0.7228	0.00	1.0209	Right (2.124)
WP-Calypso	0.77	0.4262	0.99	Right (1.042)
Rust	0.8812	0.2422	1.2184	Right (1.574)
Django	0.7102	0.00	1.0183	Right (2.092)
Nova	0.8764	0.7219	1.0609	Right (0.437)
JGroup	0.3983	0.00	0.8194	Right (1.458)
Broadleaf	1.0388	0.5586	1.3546	Right (1.064)

Table C2
Summary statistics of *entropy* across all 13 project datasets.

Project	Mean	Median	SD	Skew (Pearson's 2nd coeff.)
Brackets	365.39	143.00	511.38	Right (1.305)
Camel	5149.49	844.00	8191.84	Right (1.577)
CoreFX	3538.41	440.00	8407.69	Right (1.106)
Fabric 8	895.08	179.00	1469.96	Right (1.462)
Rails	428.39	77.0	746.29	Right (1.413)
TensorFlow	20180.28	596.5	34515.36	Right (1.702)
VSCode	4739.20	2312.5	5942.17	Right (1.225)
WP-Calypso	507.74	206.0	778.86	Right (1.162)
Rust	3835.32	673.5	6940.66	Right (1.367)
Django	802.20	268.00	1432.75	Right (1.119)
Nova	363.62	90.0	744.30	Right (1.103)
JGroup	3342.80	2539.0	3167.17	Right (0.761)
Broadleaf	1643.81	327.50	3920.65	Right (1.007)

Table C3
Summary statistics of *subsystem experience* across all 13 project datasets.

Project	Mean	Median	SD	Skew (Pearson's 2nd coeff.)
Brackets	43.76	1.61	264.21	Right (0.479)
Camel	63.45	1.06	682.88	Right (0.274)
CoreFX	25.11	1.07	228.37	Right (0.316)
Fabric 8	186.125	1.20	3249.38	Right (0.171)
Rails	65.09	1.119	549.86	Right (0.349)
TensorFlow	17.61	1.13	246.65	Right (0.201)
VSCode	52.53	1.40	422.99	Right (0.363)
WP-Calypso	48.91	1.11	337.92	Right (0.424)
Rust	74.79	1.13	1321.50	Right (0.167)
Django	41.75	1.03	414.70	Right (0.295)
Nova	124.65	1.04	2619.02	Right (0.142)
JGroup	47.87	1.0039	1232.16	Right (0.114)
Broadleaf	129.34	1.09	879.91	Right (0.437)

Table C4
Summary statistics of *recent experience* across all 13 project datasets.

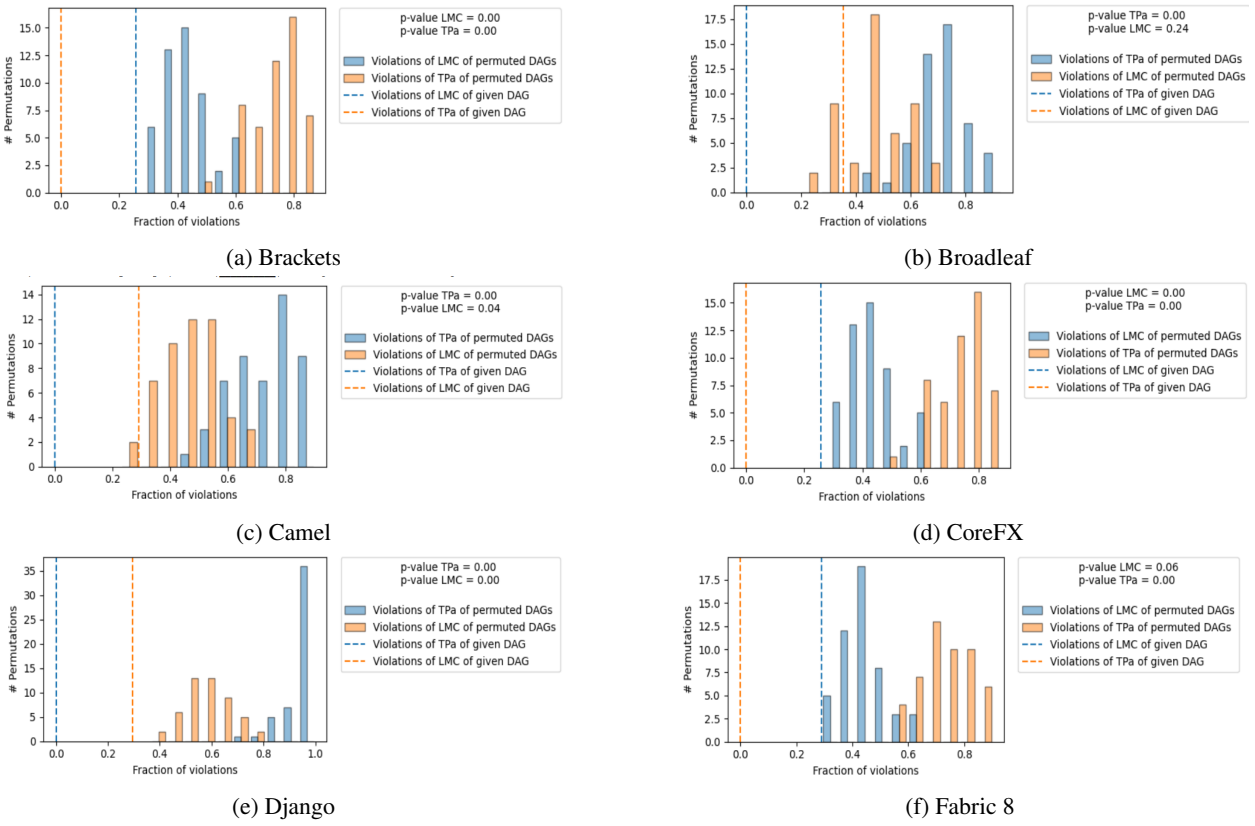
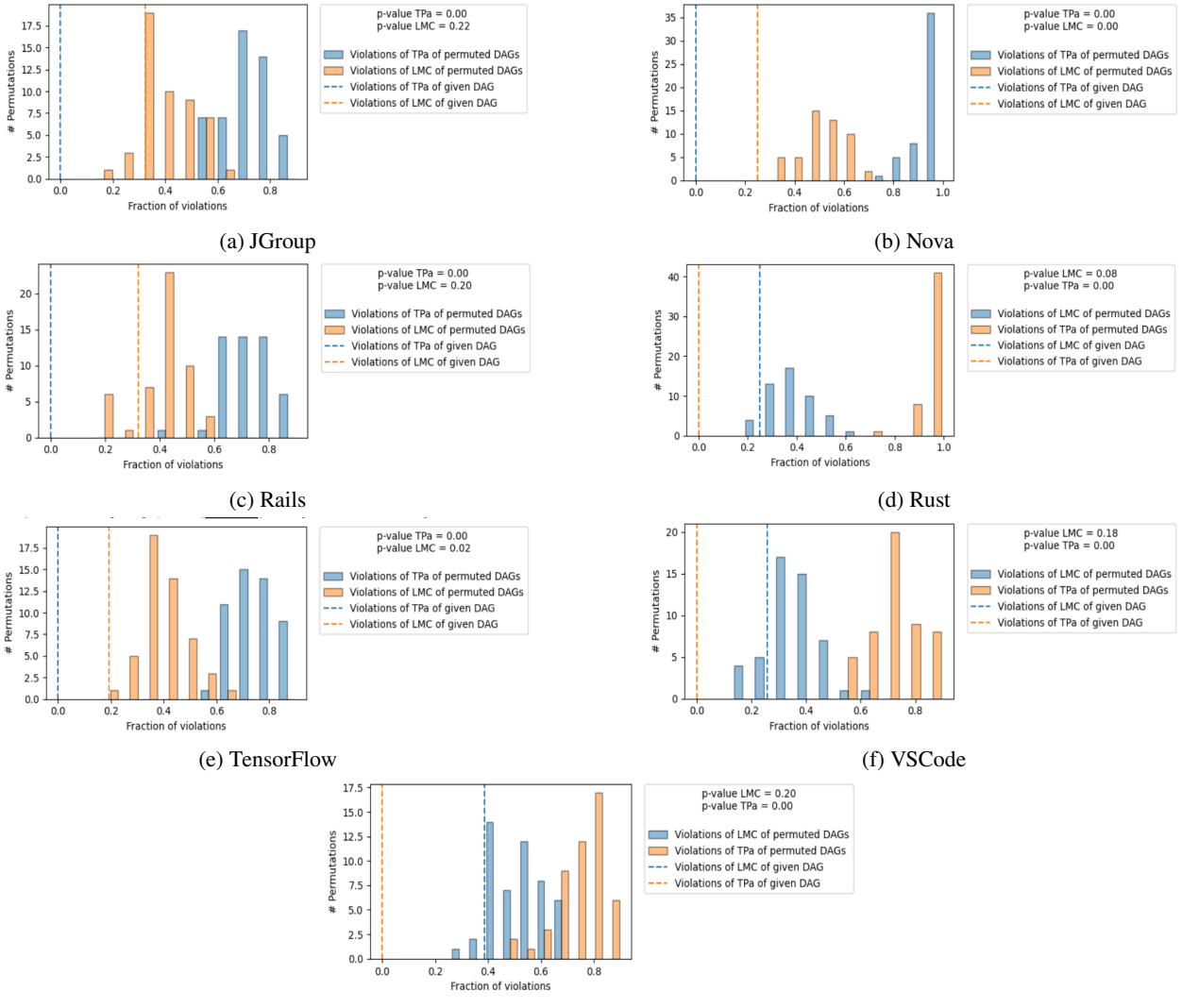


Figure D1: DAG falsification summary of (a) Brackets, (b) Broadleaf, (c) Camel, (d) CoreFX, (e) Django, and (f) Fabric 8.

Table E1
Causal effect of `ndev` on defect lifetime: days per additional developer (`ndev`) across the 11 projects compatible with the causal graph. Estimates are back-door-adjusted linear regressions on the adjustment set {time, `lt`, `nuc`}, fitted to defect-inducing commits after removal of the right-censored tail; n is the number of commits analysed. 95% CIs and p-values use HC3 heteroskedasticity-robust standard errors; bold p-values are below 0.05. RCC: estimate after adding a simulated random common cause; DSR: mean estimate over random 80% subsamples; PT: estimate after replacing the treatment with a permuted placebo (50 simulations each; no refuter rejected any estimate, all refuter $p \geq 0.31$).

		Estimate			Refutation		
Project	n	Effect	95% CI	p	RCC	DSR	PT
Brackets	3,860	−0.0682	[−1.501, 1.365]	0.926	−0.0676	−0.0275	−0.00121
Camel	5,471	−2.408	[−3.064, −1.751]	**<0.001**	−2.406	−2.420	0.0221
CoreFX	1,286	−0.312	[−0.538, −0.0863]	**0.007**	−0.312	−0.323	−0.0244
Django	10,510	0.174	[−0.160, 0.508]	0.308	0.174	0.189	0.000700
Fabric8	2,561	1.556	[0.556, 2.555]	**0.002**	1.555	1.565	−0.0523
Nova	11,011	0.596	[0.381, 0.810]	**<0.001**	0.596	0.593	0.000937
Rails	13,086	−0.900	[−1.169, −0.630]	**<0.001**	−0.900	−0.903	0.00836
Rust	1,422	−0.427	[−0.734, −0.120]	**0.006**	−0.427	−0.416	−0.00673
TensorFlow	12,926	−0.249	[−0.345, −0.153]	**<0.001**	−0.249	−0.255	−0.00244
VSCode	1,023	−1.097	[−3.568, 1.373]	0.384	−1.094	−1.035	0.0578
WP-Calypso	5,596	−0.865	[−1.170, −0.561]	**<0.001**	−0.865	−0.855	0.0108



(a) JGroup

(b) Nova

(c) Rails

(d) Rust

(e) TensorFlow

(f) VSCode

(g) WP-Calypso

Figure D2: DAG falsification summary of (a) JGroup, (b) Nova, (c) Rails, (d) Rust, (e) TensorFlow, (f) VSCode and (g) WP-Calypso.

Table E2
Causal effect of `entropy` on defect lifetime: days per unit of entropy (`entropy`) across the 11 projects compatible with the causal graph. Estimates are back-door-adjusted linear regressions on the adjustment set {`nf`, `la`, `nuc`}, fitted to defect-inducing commits after removal of the right-censored tail; n is the number of commits analysed. 95% CIs and p-values use HC3 heteroskedasticity-robust standard errors; bold p-values are below 0.05. RCC: estimate after adding a simulated random common cause; DSR: mean estimate over random 80% subsamples; PT: estimate after replacing the treatment with a permuted placebo (50 simulations each; no refuter rejected any estimate, all refuter $p \geq 0.31$).

		Estimate			Refutation		
Project	n	Effect	95% CI	p	RCC	DSR	PT
Brackets	3,860	5.92	[−16.60, 28.44]	0.606	5.89	5.14	0.16
Camel	5,471	−4.26	[−15.89, 7.37]	0.473	−4.26	−4.07	0.82
CoreFX	1,286	−10.43	[−20.84, −0.0056]	**0.0499**	−10.41	−9.28	−0.11
Django	10,510	−137.55	[−149.34, −125.75]	**<0.001**	−137.54	−138.01	−1.11
Fabric8	2,561	−10.08	[−15.55, −4.61]	**<0.001**	−10.07	−10.22	−0.056
Nova	11,011	−3.58	[−9.69, 2.54]	0.251	−3.58	−3.78	−0.17
Rails	13,086	−38.06	[−54.18, −21.93]	**<0.001**	−38.05	−38.99	−1.23
Rust	1,422	−1.70	[−14.45, 11.05]	0.794	−1.68	−1.42	0.11
TensorFlow	12,926	−14.43	[−17.65, −11.22]	**<0.001**	−14.44	−14.45	−0.0055
VSCode	1,023	−8.85	[−20.59, 2.89]	0.139	−8.85	−8.46	0.79
WP-Calypso	5,596	−31.83	[−40.45, −23.22]	**<0.001**	−31.85	−31.10	−0.0012

Table E3
Causal effect of `sexp` on defect lifetime: days per additional prior change to the subsystem (`sexp`) across the 11 projects compatible with the causal graph. Estimates are back-door-adjusted linear regressions on the adjustment set {`author_time`, `ns`}, fitted to defect-inducing commits after removal of the right-censored tail; n is the number of commits analysed. 95% CIs and p-values use HC3 heteroskedasticity-robust standard errors; bold p-values are below 0.05. RCC: estimate after adding a simulated random common cause; DSR: mean estimate over random 80% subsamples; PT: estimate after replacing the treatment with a permuted placebo (50 simulations each; no refuter rejected any estimate, all refuter $p \geq 0.31$).

		Estimate			Refutation		
Project	n	Effect	95% CI	p	RCC	DSR	PT
Brackets	3,860	0.0116	[−0.0126, 0.0357]	0.348	0.0116	0.0118	−0.00260
Camel	5,471	0.00444	[0.00279, 0.00609]	**<0.001**	0.00444	0.00436	0.0000401
CoreFX	1,286	0.00159	[−0.00128, 0.00446]	0.278	0.00159	0.00155	0.0000792
Django	10,510	0.00957	[0.000403, 0.0187]	**0.041**	0.00957	0.0101	−0.000310
Fabric8	2,561	−0.0152	[−0.0190, −0.0113]	**<0.001**	−0.0152	−0.0152	0.0000289
Nova	11,011	−0.00555	[−0.0143, 0.00325]	0.216	−0.00554	−0.00527	−0.000230
Rails	13,086	0.0107	[−0.00330, 0.0247]	0.134	0.0107	0.0110	−0.00209
Rust	1,422	−0.000327	[−0.00269, 0.00204]	0.786	−0.000332	−0.000298	−0.000268
TensorFlow	12,926	0.000289	[0.000201, 0.000376]	**<0.001**	0.000289	0.000283	-1.222×10^{-6}
VSCode	1,023	−0.000933	[−0.00343, 0.00157]	0.465	−0.000936	−0.000844	0.000113
WP-Calypso	5,596	0.00397	[−0.00531, 0.0133]	0.402	0.00397	0.00380	0.000306

Table E4
Causal effect of `rexp` on defect lifetime: days per unit of recent experience (`rexp`) across the 11 projects compatible with the causal graph. Estimates are back-door-adjusted linear regressions on the adjustment set {author_time, exp}, fitted to defect-inducing commits after removal of the right-censored tail; n is the number of commits analysed. 95% CIs and p-values use HC3 heteroskedasticity-robust standard errors; bold p-values are below 0.05. RCC: estimate after adding a simulated random common cause; DSR: mean estimate over random 80% subsamples; PT: estimate after replacing the treatment with a permuted placebo (50 simulations each; no refuter rejected any estimate, all refuter $p \geq 0.31$).

		Estimate			Refutation		
Project	n	Effect	95% CI	p	RCC	DSR	PT
Brackets	3,860	−0.138	[−0.224, −0.0527]	**0.002**	−0.138	−0.139	0.00791
Camel	5,471	−0.0203	[−0.0310, −0.00968]	**<0.001**	−0.0204	−0.0200	−0.00241
CoreFX	1,286	0.299	[−0.292, 0.890]	0.321	0.299	0.253	−0.0215
Django	10,510	0.0116	[−0.0196, 0.0428]	0.468	0.0116	0.0119	−0.00146
Fabric8	2,561	0.0408	[−0.0754, 0.157]	0.492	0.0408	0.0360	−0.00343
Nova	11,011	0.0000126	[−0.00111, 0.00113]	0.982	0.0000130	9.741×10^{-6}	−0.000179
Rails	13,086	−0.00230	[−0.0199, 0.0153]	0.798	−0.00229	−0.00159	−0.00165
Rust	1,422	0.00285	[−0.0241, 0.0298]	0.836	0.00287	0.00389	0.000773
TensorFlow	12,926	−0.00587	[−0.0215, 0.00978]	0.462	−0.00588	−0.00668	−0.00100
VSCode	1,023	−0.0128	[−0.0805, 0.0549]	0.711	−0.0128	−0.00820	0.00583
WP-Calypso	5,596	0.00978	[−0.0257, 0.0453]	0.589	0.00976	0.00951	−0.00221